

101. If $a = \lim_{x \rightarrow 0} \frac{\sqrt{1+\sqrt{1+x^4}} - \sqrt{2}}{x^4}$ and $b = \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1+\cos x}}$, then the value of ab^3 is :

यदि If $a = \lim_{x \rightarrow 0} \frac{\sqrt{1+\sqrt{1+x^4}} - \sqrt{2}}{x^4}$ तथा $b = \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1+\cos x}}$ हैं, तो ab^3 का मान है :

(A) 36

(B) 25

(C) 32

(D) 30

Ans. C

Sol.
$$a = \lim_{x \rightarrow 0} \frac{\sqrt{1+\sqrt{1+x^4}} - \sqrt{2}}{x^4}$$

$$= \lim_{x \rightarrow 0} \frac{\sqrt{1+x^4} - 1}{x^4 \left(\sqrt{1+\sqrt{1+x^4}} + \sqrt{2} \right)}$$

$$= \lim_{x \rightarrow 0} \frac{x^4}{x^4 \left(\sqrt{1+\sqrt{1+x^4}} + \sqrt{2} \right) \left(\sqrt{1+x^4} + \right)}$$

Applying limit $a = \frac{1}{4\sqrt{2}}$

$$b = \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1+\cos x}}$$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos x)(\sqrt{2} + \sqrt{1+\cos x})}{2 - (1 + \cos x)}$$

$$b = \lim_{x \rightarrow 0} (1 + \cos x)(\sqrt{2} + \sqrt{1+\cos x})$$

Applying limit $b = 2(\sqrt{2} + \sqrt{2}) = 4\sqrt{2}$

Now, $ab^3 = \frac{1}{4\sqrt{2}} \times (4\sqrt{2})^3 = 32$

102. $\lim_{x \rightarrow 0} \frac{e^{2|\sin x|} - 2|\sin x| - 1}{x^2} =$

(A) does not exist

(B) is equal to 1

(C) is equal to 2

(D) is equal to -1

$$\lim_{x \rightarrow 0} \frac{e^{2|\sin x|} - 2|\sin x| - 1}{x^2} :$$

(A) का अस्तित्व नहीं है

(B) 1 के बराबर है

(C) 2 के बराबर है

(D) -1 के बराबर है

Ans. C

Sol.
$$\lim_{x \rightarrow 0} \frac{e^{2|\sin x|} - 2|\sin x| - 1}{x^2}$$

Since, it is an even function
 $\Rightarrow$ LHL = RHL

$$\text{RHL} = \lim_{x \rightarrow 0^+} \frac{e^{2\sin x} - 2\sin x - 1}{x^2}$$

$$= \lim_{x \rightarrow 0^+} \frac{e^{2\sin x} (2\cos x) - 2\cos x}{2x}$$

$$= \lim_{x \rightarrow 0^+} = \left(\frac{e^{2\sin x} - 1}{2\sin x} \right) \left(\frac{2\sin x}{x} \right) \cos x$$

$$= 1 \times 2 \times 1$$

$$= 2$$

103. Let $f : \mathbb{R} \rightarrow (0, \infty)$ be strictly increasing function such that $\lim_{x \rightarrow \infty} \frac{f(7x)}{f(x)} = 1$. Then, the value of $\lim_{x \rightarrow \infty} \left[\frac{f(5x)}{f(x)} - 1 \right]$ is equal to :

माना $f : \mathbb{R} \rightarrow (0, \infty)$ निरंतर वर्धमान फलन है तथा $\lim_{x \rightarrow \infty} \frac{f(7x)}{f(x)} = 1$ है। तो $\lim_{x \rightarrow \infty} \left[\frac{f(5x)}{f(x)} - 1 \right]$ का मान बराबर है :

(A) $\frac{7}{5}$

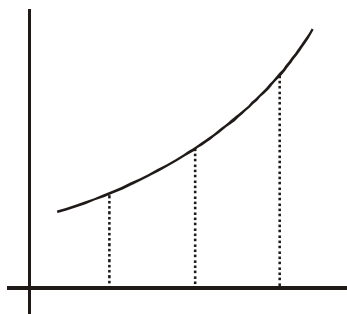
(B) 0

(C) 1

(D) 4

Ans. B

Sol. $f(x) \leq f(5x) \leq f(7x), \forall x > 0$
 $x < 5x < 7x$



$$1 \leq \frac{f(5x)}{f(x)} \leq \frac{f(7x)}{f(x)}$$

$$\text{As } x \rightarrow \infty, \frac{f(5x)}{f(x)} \rightarrow 1^+$$

$$\lim_{x \rightarrow \infty} \left[\frac{f(5x)}{f(x)} - 1 \right] = 0$$

104. Let $f(x) = \begin{cases} x-1, & x \text{ is even,} \\ 2x, & x \text{ is odd,} \end{cases} x \in \mathbb{N}$. If for some $a \in \mathbb{N}$, $f(f(f(a))) = 21$, then $\lim_{x \rightarrow a^-} \left\{ \frac{|x|^3}{a} - \left[\frac{x}{a} \right] \right\}$, where $[t]$ denotes the greatest integer less than or equal to t , is equal to :

माना $f(x) = \begin{cases} x-1, & x \text{ is even,} \\ 2x, & x \text{ is odd,} \end{cases} x \in \mathbb{N}$ यदि किसी $a \in \mathbb{N}$ के लिए $f(f(f(a))) = 21$, तो $\lim_{x \rightarrow a^-} \left\{ \frac{|x|^3}{a} - \left[\frac{x}{a} \right] \right\}$, जहाँ $[t]$ के

सम्मान या कम महत्तम पूर्णांक दर्शाता है, बराबर होगा :

(A) 225

(B) 144

(C) 121

(D) 169

Ans. B

Sol. $f(f(f(a))) = 21$

$$f(f(a)) - 1 = 21$$

$$f(f(a)) = 22$$

$$2f(a) = 22$$

$$f(a) = 11$$

$$a - 1 = 11$$

$$\Rightarrow a = 12$$

$$\lim_{x \rightarrow 12^-} \left(\frac{|x|^3}{12} - \left[\frac{x}{12} \right] \right)$$

$$= 144 - 0 = 144$$

105. Let $a_1, a_2, a_3, \dots, a_n$ be n positive consecutive terms of an arithmetic progression. if $d > 0$ is its common difference,

then $\lim_{n \rightarrow \infty} \sqrt{\frac{d}{n}} \left(\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}} \right)$ is :

माना $a_1, a_2, a_3, \dots, a_n$ समांतर श्रेणी के धनात्मक क्रमागत n पद हैं। यदि सार्वन्तर $d > 0$ है, तब

$\lim_{n \rightarrow \infty} \sqrt{\frac{d}{n}} \left(\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}} \right)$ का मान है :

- (A) 1 (B) 0 (C) $\sqrt{d}$ (D) $\frac{1}{\sqrt{d}}$

Ans. A

Sol.

$$\begin{aligned} & \left(\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}} \right) \\ &= \left(\frac{\sqrt{a_2} - \sqrt{a_1}}{a_2 - a_1} + \frac{\sqrt{a_3} - \sqrt{a_2}}{a_3 - a_2} + \frac{\sqrt{a_4} - \sqrt{a_3}}{a_4 - a_3} + \dots + \frac{\sqrt{a_n} - \sqrt{a_{n-1}}}{a_n - a_{n-1}} \right) \\ &= \frac{1}{d} (\sqrt{a_n} - \sqrt{a_1}) \\ &\therefore \lim_{n \rightarrow \infty} \sqrt{\frac{d}{n}} \left(\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}} \right) \\ &= \lim_{n \rightarrow \infty} \sqrt{\frac{d}{n}} \left(\frac{\sqrt{a_n} - \sqrt{a_1}}{d} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{\sqrt{a_1 + (n-1)d} - \sqrt{a_1}}{\sqrt{n} \sqrt{d}} \right) = \lim_{n \rightarrow \infty} \left(\sqrt{\frac{a_1}{nd}} + \left(1 - \frac{1}{n} \right) - \sqrt{\frac{a_1}{nd}} \right) = 1 \end{aligned}$$

106. $\lim_{n \rightarrow \infty} \left\{ \left(2^{\frac{1}{2}} - 2^{\frac{1}{3}} \right) \left(2^{\frac{1}{2}} - 2^{\frac{1}{5}} \right) \dots \left(2^{\frac{1}{2}} - 2^{\frac{1}{2n+1}} \right) \right\}$ is equal to :

$\lim_{n \rightarrow \infty} \left\{ \left(2^{\frac{1}{2}} - 2^{\frac{1}{3}} \right) \left(2^{\frac{1}{2}} - 2^{\frac{1}{5}} \right) \dots \left(2^{\frac{1}{2}} - 2^{\frac{1}{2n+1}} \right) \right\}$ बराबर है :

- (A) 1 (B) $\sqrt{2}$ (C) 0 (D) $\frac{1}{\sqrt{2}}$

Ans. C

Sol.
$$P = \lim_{n \rightarrow \infty} \left(2^{\frac{1}{2}} - 2^{\frac{1}{3}} \right) \left(2^{\frac{1}{2}} - 2^{\frac{1}{5}} \right) \dots \left(2^{\frac{1}{2}} - 2^{\frac{1}{2n+1}} \right)$$

Let

$$2^{\frac{1}{2}} - 2^{\frac{1}{3}} \rightarrow \text{Smallest}$$

$$2^{\frac{1}{2}} - 2^{\frac{1}{2n+1}} \rightarrow \text{Largest}$$

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$$\left(2^{\frac{1}{2}} - 2^{\frac{1}{3}} \right)^n \leq P \leq \left(2^{\frac{1}{2}} - 2^{\frac{1}{2n+1}} \right)^n$$

$$\left(\begin{array}{c} \text{lie b/w} \\ 0 \text{ and } 1 \end{array} \right)^n$$

$$\lim_{n \rightarrow \infty} \left(2^{\frac{1}{2}} - 2^{\frac{1}{3}} \right)^n = 0$$

$$\lim_{n \rightarrow \infty} \left(2^{\frac{1}{2}} - 2^{\frac{1}{2n+1}} \right)^n = 0$$

$$\therefore P = 0$$

107. For $t > -1$, let α_t and β_t be the roots of the equation $\left((t+2)^{\frac{1}{7}} - 1 \right) x^2 + \left((t+2)^{\frac{1}{6}} - 1 \right) x + \left((t+2)^{\frac{1}{21}} - 1 \right) = 0$. If

$\lim_{t \rightarrow -1^+} \alpha_t = a$ and $\lim_{t \rightarrow -1^+} \beta_t = b$, then $72(a+b)^2$ is equal to _____.

माना $t > -1$ के लिए समीकरण $\left((t+2)^{\frac{1}{7}} - 1 \right) x^2 + \left((t+2)^{\frac{1}{6}} - 1 \right) x + \left((t+2)^{\frac{1}{21}} - 1 \right) = 0$ के मूल α_t और β_t हैं। यदि

$\lim_{t \rightarrow -1^+} \alpha_t = a$ और $\lim_{t \rightarrow -1^+} \beta_t = b$ हैं, तो $72(a+b)^2$ बराबर है _____.

Ans. 98

Sol.
$$\alpha_t + \beta_t = \frac{(t+2)^{\frac{1}{6}} - 1}{(t+2)^{\frac{1}{7}} - 1}$$

$$\lim_{t \rightarrow -1^+} (\alpha_t + \beta_t) = \frac{\frac{1}{6}}{\frac{1}{7}} = \frac{7}{6}$$

$$\Rightarrow \lim_{t \rightarrow -1^+} \alpha_t + \lim_{t \rightarrow -1^+} \beta_t = \frac{7}{6}$$

$$\text{So, } a + b = \frac{7}{6}$$

$$72(a+b)^2 = 72 \cdot \frac{49}{36} = 98$$

108. Let $f(x) = \lim_{n \rightarrow \infty} \sum_{r=0}^n \left(\frac{\tan\left(\frac{x}{2^{r+1}}\right) + \tan^3\left(\frac{x}{2^{r+1}}\right)}{1 - \tan^2\left(\frac{x}{2^{r+1}}\right)} \right)$. Then $\lim_{x \rightarrow 0} \frac{e^x - e^{f(x)}}{(x - f(x))}$ is equal to _____.

माना $f(x) = \lim_{n \rightarrow \infty} \sum_{r=0}^n \left(\frac{\tan\left(\frac{x}{2^{r+1}}\right) + \tan^3\left(\frac{x}{2^{r+1}}\right)}{1 - \tan^2\left(\frac{x}{2^{r+1}}\right)} \right)$ है। तब $\lim_{x \rightarrow 0} \frac{e^x - e^{f(x)}}{(x - f(x))}$ बराबर है _____।

Ans. 1

Sol. $f(x) = \lim_{n \rightarrow \infty} \sum_{r=0}^n \left(\frac{\tan \theta + \tan^3 \theta}{1 - \tan^2 \theta} \right)$

$$\theta = \left(\frac{x}{2^{r+1}} \right)$$

$$f(x) = \sum_{r=1}^n (\tan \theta) \frac{(1 + \tan^2 \theta)}{(1 - \tan^2 \theta)}$$

$$f(x) = \sum_{r=1}^n \frac{(\tan \theta)}{(\cos 2\theta)}$$

$$f(x) = \sum_{r=1}^n \frac{\sin(2\theta - \theta)}{\cos \theta \cos 2\theta}$$

$$f(x) = \sum_{r=1}^n \left(\frac{\sin 2\theta \cdot \cos \theta - \cos 2\theta \sin \theta}{\cos \theta \cos 2\theta} \right)$$

$$f(x) = \sum_{r=1}^n (\tan 2\theta - \tan \theta)$$

$$f(x) = \sum_{r=1}^n \left[\tan\left(\frac{x}{2^r}\right) - \tan\left(\frac{x}{2^{r+1}}\right) \right]$$

$$f(x) = \tan\left(\frac{x}{2}\right) - \tan\left(\frac{x}{4}\right)$$

+

$$\tan\left(\frac{x}{4}\right) - \tan\left(\frac{x}{8}\right)$$

+

$\vdots$

$$\tan\left(\frac{x}{2^n}\right) - \tan\left(\frac{x}{2^{n+1}}\right)$$

$$f(x) = \lim_{n \rightarrow \infty} \tan\left(\frac{x}{2}\right) - \tan\left(\frac{x}{2^{n+1}}\right)$$

$$f(x) = \tan\left(\frac{x}{2}\right)$$

$$= \lim_{x \rightarrow 0} \frac{e^x - e^{\tan\left(\frac{x}{2}\right)}}{x - \tan\left(\frac{x}{2}\right)}$$

By series expansion of $e^s = 1 + s$

$$\lim_{x \rightarrow 0} \frac{(1+x) - \left(1 + \tan\left(\frac{x}{2}\right)\right)}{\left(x - \tan\left(\frac{x}{2}\right)\right)} = 1$$

109. Let $[t]$ be the greatest integer less than or equal to t . Then least value of $p \in \mathbb{N}$ for which

$$\lim_{x \rightarrow 0^+} \left(x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{p}{x} \right] \right) - x^2 \left(\left[\frac{1}{x^2} \right] + \left[\frac{2^2}{x^2} \right] + \dots + \left[\frac{9^2}{x^2} \right] \right) \right) \geq 1 \text{ is equal to } \underline{\hspace{2cm}}.$$

माना कि $[t]$, t से छोटा या उसके बराबर का महत्तम पूर्णांक है। तब $p \in \mathbb{N}$ का न्यूनतम मान, जिसके लिए

$$\lim_{x \rightarrow 0^+} \left(x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{p}{x} \right] \right) - x^2 \left(\left[\frac{1}{x^2} \right] + \left[\frac{2^2}{x^2} \right] + \dots + \left[\frac{9^2}{x^2} \right] \right) \right) \geq 1 \text{ है, बराबर है } \underline{\hspace{2cm}}.$$

Ans. 24

Sol. $\lim_{x \rightarrow 0^+} \left(x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{p}{x} \right] \right) - x^2 \left(\left[\frac{1}{x^2} \right] + \left[\frac{2^2}{x^2} \right] + \dots + \left[\frac{9^2}{x^2} \right] \right) \right) \geq 1$

$$x - 1 < [x] \leq x$$

$$\frac{1}{x} - 1 < \left\lfloor \frac{1}{x} \right\rfloor \leq \frac{1}{x}$$

$$\frac{P(P+1)}{2x} - P < \dots \leq \frac{P(P+1)}{2x}$$

$$\frac{P(P+1)}{2} - Px < \dots \leq \frac{P(P+1)}{2}$$

$$P^2 + P - 572 \geq 0$$

$$P = \frac{-1 \pm \sqrt{1 + 4 \times 572}}{2}$$

$$P = \frac{-1 \pm \sqrt{2289}}{2}$$

$$P = \frac{46.84}{2} \text{ or } -\frac{48.84}{2}$$

$$P = 23.42 \text{ or } -24.42$$

$$(P + 24.42)(P - 23.42) \geq 0$$

$$P \in (-\infty, -24.42] \cup [23.42, +\infty)$$

$$P = 24$$

110. If $\lim_{x \rightarrow 1} \frac{(5x+1)^{\frac{1}{3}} - (x+5)^{\frac{1}{3}}}{(2x+3)^{\frac{1}{2}} - (x+4)^{\frac{1}{2}}} = \frac{m\sqrt{5}}{n(2n)^{\frac{2}{3}}}$, where $\gcd(m, n) = 1$, then $8m + 12n$ is equal to _____.

यदि $\lim_{x \rightarrow 1} \frac{(5x+1)^{\frac{1}{3}} - (x+5)^{\frac{1}{3}}}{(2x+3)^{\frac{1}{2}} - (x+4)^{\frac{1}{2}}} = \frac{m\sqrt{5}}{n(2n)^{\frac{2}{3}}}$, जहाँ $\gcd(m, n) = 1$ हैं, तो $8m + 12n$ बराबर है _____

Ans. 100

Sol. $\ell = \lim_{x \rightarrow 1} \frac{(5x+1)^{\frac{1}{3}} - (x+5)^{\frac{1}{3}}}{(2x+3)^{\frac{1}{2}} - (x+4)^{\frac{1}{2}}}$

From : $\frac{0}{0}$, using L-H rule

$$\ell = \lim_{x \rightarrow 1} \frac{\frac{1}{3} \times 5(5x+1)^{-\frac{2}{3}} - \frac{1}{3}(x+5)^{-\frac{2}{3}}}{\frac{1}{2} \times 2(2x+3)^{-\frac{1}{2}} - \frac{1}{2}(x+4)^{-\frac{1}{2}}}$$

$$= \frac{\left(\frac{5}{3} - \frac{1}{3}\right) 6^{-\frac{2}{3}}}{\frac{1}{2} 5^{-\frac{1}{2}}} = \frac{8}{3} \times \frac{5^{\frac{1}{2}}}{6^{\frac{2}{3}}} = \frac{m\sqrt{5}}{n(2n)^{\frac{2}{3}}}$$

$$\Rightarrow m = 8, n = 3$$

$$\Rightarrow 8m + 12n = 100$$

111. Let $a > 0$ be a root of the equation $2x^2 + x - 2 = 0$. If $\lim_{x \rightarrow \frac{1}{a}} \frac{16(1 - \cos(2 + x - 2x^2))}{(1 - ax)^2} = \alpha + \beta\sqrt{17}$ where $\alpha, \beta \in \mathbb{Z}$, then $\alpha + \beta$ is equal to _____.

माना समीकरण $2x^2 + x - 2 = 0$ का एक मूल $a > 0$ है। यदि $\lim_{x \rightarrow \frac{1}{a}} \frac{16(1 - \cos(2 + x - 2x^2))}{(1 - ax)^2} = \alpha + \beta\sqrt{17}$ है, जहाँ $\alpha, \beta \in \mathbb{Z}$ है। तो $\alpha + \beta$ बराबर है _____

Ans. 170

Sol. $\therefore 2x^2 + x - 2 = 0$ has two roots where

$$a = \frac{\sqrt{17} - 1}{4} \text{ and another root is } \frac{-\sqrt{17} - 1}{4}$$

$$\text{And } 2 + x - 2x^2 = -2 \left(x - \frac{1}{a} \right) \left(x + \frac{4}{\sqrt{17} + 1} \right)$$

$$\text{Now } \lim_{x \rightarrow \frac{1}{a}} \frac{16(1 - \cos(2 + x - 2x^2))}{(1 - ax)^2}$$

$$\lim_{x \rightarrow \frac{1}{a}} \frac{32 \sin^2(2 + x - 2x^2)}{a^2 \left(\frac{1}{a} - x \right)^2}$$

$$= \lim_{x \rightarrow \frac{1}{a}} \frac{\left(x + \frac{4}{\sqrt{17} + 1} \right)^2 32 \left(\sin \left(\frac{1}{2}(-2) \right) \left(x - \frac{1}{a} \right) \left(x + \frac{4}{\sqrt{17} + 1} \right) \right)^2}{a^2 \left(\left(x - \frac{1}{a} \right) \left(x + \frac{4}{\sqrt{17} + 1} \right) \right)^2}$$

$$\begin{aligned}
 &= 2 \left(\frac{1}{a} + \frac{4}{\sqrt{17}+1} \right)^2 \left(\frac{4}{\sqrt{17}-1} \right)^2 \\
 &= 2 \left(\frac{4}{\sqrt{17}-1} + \frac{4}{\sqrt{17}+1} \right)^2 \left(\frac{4}{\sqrt{17}-1} \right)^2 \\
 &= \frac{17 \times 4}{18-2\sqrt{17}} = \frac{68}{9-\sqrt{17}} \\
 &= 17(9+\sqrt{17}) \\
 \alpha + \beta &= 170
 \end{aligned}$$

112. If $\alpha = \lim_{x \rightarrow 0^+} \left(\frac{e^{\sqrt{\tan x}} - e^{\sqrt{x}}}{\sqrt{\tan x} - \sqrt{x}} \right)$ and $\beta = \lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{2} \cot x}$ are the roots of the quadratic equation $ax^2 + bx - \sqrt{e} = 0$, then $12 \log_e(a+b)$ is equal to _____.

यदि $\alpha = \lim_{x \rightarrow 0^+} \left(\frac{e^{\sqrt{\tan x}} - e^{\sqrt{x}}}{\sqrt{\tan x} - \sqrt{x}} \right)$ तथा $\beta = \lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{2} \cot x}$ द्विघात समीकरण $ax^2 + bx - \sqrt{e} = 0$ के मूल हैं, तो

$12 \log_e(a+b)$ बराबर है _____.

Ans. 6

Sol.

$$\begin{aligned}
 \alpha &= \lim_{x \rightarrow 0^+} \frac{e^{\sqrt{\tan x}} - e^{\sqrt{x}}}{\sqrt{\tan x} - \sqrt{x}} \\
 &= \lim_{x \rightarrow 0} \frac{e^{\sqrt{x}} (e^{\sqrt{\tan x} - \sqrt{x}} - 1)}{(\sqrt{\tan x} - \sqrt{x})} = 1
 \end{aligned}$$

$$\begin{aligned}
 \beta &= \lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{2} \cot x} = \lim_{x \rightarrow 0} e^{(\sin x) \left(\frac{1}{2} \cot x \right)} \\
 &= \lim_{x \rightarrow 0} e^{\frac{1}{2} \cos x} = e^{\frac{1}{2}}
 \end{aligned}$$

$$\text{Product of roots} = \sqrt{e} = \frac{-\sqrt{e}}{a} \Rightarrow a = -1$$

$$\text{Sum of roots} = \frac{-b}{a} = 1 + \sqrt{e}$$

$$= b = \sqrt{e} + 1$$

$$\Rightarrow 12 \ln(a+b) = 12 \ln(\sqrt{e} + 1 - 1) = 12 \ln\left(e^{\frac{1}{2}}\right) = 6$$

Continuity & Differentiability

113. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying $f(0) = 1$ and $f(2x) - f(x) = x$ for all $x \in \mathbb{R}$. If

$$\lim_{n \rightarrow \infty} \left\{ f(x) - f\left(\frac{x}{2^n}\right) \right\} = G(x), \text{ then } \sum_{r=1}^{10} G(r^2) \text{ is equal to :}$$

माना एक संतत फलन $f : \mathbb{R} \rightarrow \mathbb{R}$ के लिए $f(0) = 1$ और $f(2x) - f(x) = x, x \in \mathbb{R}$ हैं। यदि $\lim_{n \rightarrow \infty} \left\{ f(x) - f\left(\frac{x}{2^n}\right) \right\} = G(x)$ है, तो

$$\sum_{r=1}^{10} G(r^2) \text{ बराबर है :}$$

(A) 215

(B) 385

(C) 420

(D) 540

Ans. B

Sol. $f : \mathbb{R} \rightarrow \mathbb{R}$

$$f(0) = 1$$

$$f(2x) - f(x) = x \quad \forall x \in \mathbb{R}$$

$$\lim_{n \rightarrow \infty} \left\{ f(x) - f\left(\frac{x}{2^n}\right) \right\} = G(x)$$

$$\text{then } \sum_{r=1}^{10} G(r^2) = ??$$

$$\text{Now, } f(2x) - f(x) = x$$

$$f(x) - f\left(\frac{x}{2}\right) = \frac{x}{2}$$

$$f\left(\frac{x}{2}\right) - f\left(\frac{x}{2^2}\right) = \frac{x}{2^2}$$

$\vdots$

$$f\left(\frac{x}{2^{n-1}}\right) - f\left(\frac{x}{2^n}\right) = \frac{x}{2^n}$$

$$f(2x) - f\left(\frac{x}{2^n}\right) = x \left\{ 1 + \frac{1}{2} + \frac{1}{2^2} + \dots \right\}$$

$$n \rightarrow \infty$$

$$f(2x) - f(0) = x - \frac{1}{1 - \frac{1}{2}} \Rightarrow 2x$$

$$f(2x) = 2x + 1$$

$$\therefore f(x) = x + 1$$

$$\text{Now, } G(x) = f(x) - f(0)$$

$$= x + 1 - 1$$

$$= x$$

$$\text{Now, } \sum_{r=1}^{10} G(r^2) = \sum_{r=1}^{10} r^2$$

$$\Rightarrow 1^2 + 2^2 + \dots + 10^2$$

$$\Rightarrow \frac{n(n+1)(2n+1)}{6} \quad (n=10)$$

$$\Rightarrow \frac{10 \times 11 \times 21}{6}$$

$$\Rightarrow 5 \times 77$$

$$\Rightarrow 385$$

114. Let P be the foot of the perpendicular from point (1, 2, 2) on the line $L: \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2}$. Let the

line $\vec{r} = (-\hat{i} + \hat{j} - 2\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k}), \lambda \in \mathbb{R}$, intersect the line L, at Q. Then $2(PQ)^2$ is equal to :

माना कि P, बिन्दु (1, 2, 2) से रेखा $L: \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2}$ पर लम्ब का पाद है। माना कि रेखा

$\vec{r} = (-\hat{i} + \hat{j} - 2\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k}), \lambda \in \mathbb{R}$, रेखा L को Q पर काटती है। तब $2(PQ)^2$ बराबर है :

(A) 19

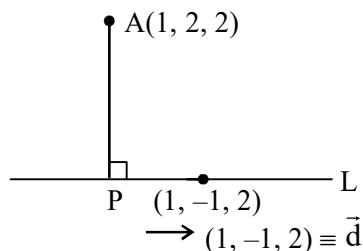
(B) 27

(C) 25

(D) 29

Ans. B

Sol.



$$L: \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2} = \mu$$

$$= \mu P(\mu+1, -\mu-1, 2\mu+2)$$

$$\overrightarrow{AP} \cdot \vec{d} = 0 \Rightarrow (\mu, -\mu-3, 2\mu) \cdot (1, -1, 2) = 0$$

$$\Rightarrow \mu + \mu + 3 + 4\mu = 0 \Rightarrow \mu = -\frac{1}{2}$$

$$\therefore P\left(\frac{-1}{2}+1, +\frac{1}{2}-1, 2\left(\frac{-1}{2}\right)+2\right)$$

$$P\left(\frac{1}{2}, \frac{-1}{2}, 1\right)$$

Now general pt. on L_2 is $Q(-1+\lambda, 1-\lambda, -2+\lambda)$

Equate it with general pt of L

$$\begin{array}{l|l|l} \mu+1=-1+\lambda & -\mu-1=1-\lambda & 2\mu+2=-2+\lambda \\ \mu=\lambda-2 & \mu=\lambda-2 & \downarrow \\ & & 2(\lambda-2)+2=-2+\lambda \\ & & 2\lambda-4+2=-2+\lambda \end{array}$$

$$\therefore \mu = -2, \lambda = 0$$

$$\therefore Q = -2, \lambda = 0$$

$$P\left(\frac{1}{2}, \frac{-1}{2}, 1\right) \text{ and } Q(-1, 1, -2)$$

$$PQ = \sqrt{\left(\frac{1}{2}+1\right)^2 + \left(\frac{-1}{2}-1\right)^2 + (1+2)^2}$$

$$= \sqrt{\frac{9}{4} + \frac{9}{4} + 9} = \sqrt{\frac{54}{4}}$$

$$\therefore 2(PQ)^2 = 2\left(\frac{54}{4}\right) = 27$$

115. Let a straight line L pass through the point $P(2, -1, 3)$ and be perpendicular to the lines $\frac{x-1}{2} = \frac{y+1}{1} = \frac{z-3}{-2}$ and

$\frac{x-3}{1} = \frac{y-2}{3} = \frac{z+2}{4}$. If the line L intersects the yz -plane at the point Q , then the distance between the points P

and Q is :

माना कि एक सरल रेखा L बिन्दु $P(2, -1, 3)$ से होकर जाती है और और रेखाओं $\frac{x-1}{2} = \frac{y+1}{1} = \frac{z-3}{-2}$ और

$\frac{x-3}{1} = \frac{y-2}{3} = \frac{z+2}{4}$ के लम्बवत् है। यदि रेखा L , yz - तल को बिन्दु Q पर काटती है, तो बिन्दुओं P और Q के बीच की दूरी है :

(A) $2\sqrt{3}$

(B) 2

(C) 3

(D) $\sqrt{10}$

Ans. C

Sol. Vector parallel to 'L'

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 3 & 4 \end{vmatrix} = 10\hat{i} - 10\hat{j} + 5\hat{k}$$

$$= 5(2\hat{i} - 2\hat{j} + \hat{k})$$

Equation of 'L' $\frac{x-2}{2} = \frac{y+1}{-2} = \frac{z-3}{1} = \lambda$ (say)

Let $Q(2\lambda + 2, -2\lambda - 1, \lambda + 3)$

$$\Rightarrow 2\lambda + 2 = 0 \quad \Rightarrow \lambda = -1$$

$$\Rightarrow Q(0, 1, 2)$$

$$d(P, Q) = 3$$

116. Let α, β ($\alpha \neq \beta$) be the values of m , for which the equations $x + y + z = 1$; $x + 2y + 4z = m$ and $x + 4y + 10z = m^2$ have infinitely many solutions. Then the value of $\sum_{n=1}^{10} (n^\alpha + n^\beta)$ is equal to :

माना कि α, β ($\alpha \neq \beta$), m के वे मान हैं, जिनके लिए समीकरणों $x + y + z = 1$; $x + 2y + 4z = m$ और

$x + 4y + 10z = m^2$ के अनन्त हल हैं। तब $\sum_{n=1}^{10} (n^\alpha + n^\beta)$ का मान बराबर है :

(A) 440

(B) 3080

(C) 3410

(D) 560

Ans. A

Sol. $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{vmatrix} = 1(20 - 16) - 1(10 - 4) + 1(4 - 2)$

$$= 4 - 6 + 2 = 0$$

For infinite solutions

$$\Delta_x = \Delta_y = \Delta_z = 0$$

$$m^2 - 3x + 2 = 0$$

$$m = 1, 2$$

$$\alpha = 1, \beta = 2$$

$$\begin{aligned}\therefore \sum_{n=1}^{10} (n^{\alpha} + n^{\beta}) &= \sum_{n=1}^{10} n^1 + \sum_{n=1}^{10} n^2 \\ &= \frac{10(11)}{2} + \frac{10(11)(21)}{6} \\ &= 55 + 385 \\ &= 440\end{aligned}$$

117. Let $f : [-1, 2] \rightarrow \mathbb{R}$ be given by $f(x) = 2x^2 + x + [x^2] - [x]$, where $[t]$ denotes the greatest integer less than or equal to t . The number of points, where f is not continuous is :

माना फलन $f : [-1, 2] \rightarrow \mathbb{R}$, $f(x) = 2x^2 + x + [x^2] - [x]$ द्वारा दिया गया है, जहाँ $[t]$ महत्तम पूर्णांक $\leq t$ हैं। उन बिन्दुओं जिन पर f संतत नहीं है, की संख्या है) :

- (A) 3 (B) 5 (C) 6 (D) 4

Ans. D

Sol. Doubtful points : $-1, 0, 1, \sqrt{2}, \sqrt{3}, 2$

at $x = \sqrt{2}, \sqrt{3}$

$f(x) = (2x^2 + x - [x]) + [x^2] = \text{Discontinuous}$

$\downarrow$ $\downarrow$
 Cont. Discont.

at $x = -1$:

$$\left. \begin{aligned} \text{RHL} \Rightarrow f(x) &= (2 - 1 - (-1)) + 0 = 2 \\ F(-1) &= 2 - 1 - (-1) + 1 = 3 \end{aligned} \right\} \text{Dis.}$$

at $x = 2$:

$$\left. \begin{aligned} \text{LHL} \Rightarrow f(x) &= 8 + 2 - 1 + 3 = 12 \\ F(2) &= 8 + 2 - 2 + 4 = 12 \end{aligned} \right\} \text{Cont.}$$

at $x = 0$:

$$\left. \begin{aligned} \text{LHL} \Rightarrow 0 + 0 - (-1) + 0 &= 1 \\ F(0) &= 0 \end{aligned} \right\} \text{Dis.}$$

at $x = 1$:

$$\left. \begin{aligned} \text{LHL} \Rightarrow 2 + 1 - 0 + 0 &= 3 \\ F(1) &= 3 - 1 + 1 = 3 \\ \text{RHL} \Rightarrow 2 + 1 - 1 + 1 &= 3 \end{aligned} \right\} \text{Cont.}$$

118. For $a, b > 0$, let $f(x) = \begin{cases} \frac{\tan((a+1)x) + b \tan x}{x}, & x < 0 \\ 3, & x = 0 \\ \frac{\sqrt{ax + b^2x^2} - \sqrt{ax}}{b\sqrt{ax}\sqrt{x}}, & x > 0 \end{cases}$ be a continuous function at $x = 0$. Then $\frac{b}{a}$ is equal to

:

$a, b > 0$ के लिए, माना $f(x) = \begin{cases} \frac{\tan((a+1)x) + b \tan x}{x}, & x < 0 \\ 3, & x = 0 \\ \frac{\sqrt{ax + b^2x^2} - \sqrt{ax}}{b\sqrt{ax}\sqrt{x}}, & x > 0 \end{cases}$ $x = 0$ पर एक संतत फलन है। तो $\frac{b}{a}$ बराबर है :

(A) 5

(B) 6

(C) 4

(D) 8

Ans. B

Sol. $\lim_{x \rightarrow 0} f(x) = f(0) = 3$

$$= \lim_{x \rightarrow 0^+} \frac{\sqrt{ax + b^2x^2} - \sqrt{ax}}{b\sqrt{ax}\sqrt{x}} = 3$$

$$= \lim_{x \rightarrow 0^+} \frac{ax + b^2x^2 - ax}{b\sqrt{ax}^{\frac{3}{2}} (\sqrt{ax + b^2x^2} + \sqrt{ax})}$$

$$= \lim_{x \rightarrow 0^+} \frac{b^2}{b\sqrt{a} (\sqrt{a + b^2x} + \sqrt{a})}$$

$$= \frac{b}{\sqrt{a} \cdot 2\sqrt{a}} \Rightarrow \frac{b}{2a} = 3 \quad \dots(1)$$

$$= \lim_{x \rightarrow 0^-} \frac{\tan(a+1)x + b \tan x}{x}$$

$$= \lim_{x \rightarrow 0^-} \frac{\frac{\tan(a+1)x}{(a+1)x} (a+1)x + \frac{b \tan x}{x} \cdot x}{x}$$

$$= a + 1 + b = 3 \Rightarrow a + b = 2$$

...(2)

from (1) and (2)

$$a + 6a = 2$$

$$a = \frac{2}{7} \quad b = \frac{12}{7}$$

119. Consider the function.

$$f(x) = \begin{cases} \frac{a(7x - 12 - x^2)}{b|x^2 - 7x + 12|} & , x < 3 \\ \frac{\sin(x-3)}{2^{x-[x]}} & , x > 3 \\ b & , x = 3 \end{cases}$$

where $[x]$ denotes the greatest integer less than or equal to x . If S denotes the set of all ordered pairs (a, b) such that $f(x)$ is continuous at $x = 3$, then the number of elements in S is :

- (A) 1 (B) 2 (C) Infinitely many (D) 4

फलन

$$f(x) = \begin{cases} \frac{a(7x - 12 - x^2)}{b|x^2 - 7x + 12|} & , x < 3 \\ \frac{\sin(x-3)}{2^{x-[x]}} & , x > 3 \\ b & , x = 3 \end{cases}$$

जहाँ $[x]$ महत्तम पूर्णांक फलन है, का विचार कीजिए। यदि सभी क्रमित युग्मों (a, b) , जिनके लिए $x = 3$ पर $f(x)$ संतत है, का समुच्चय S है, तो में अवयवों की संख्या है :

- (A) 1 (B) 2 (C) अनन्त (D) 4

Ans. A

Sol. $f(3^-) = \frac{a}{b} \frac{(7x-12-x^2)}{|x^2-7x+12|}$ (for $f(x)$ to be cont.)

$$\Rightarrow f(3^-) = \frac{-a}{b} \frac{(x-3)(x-4)}{(x-3)(x-4)}; x < 3 \Rightarrow \frac{-a}{b}$$

Hence $f(3^-) = \frac{-a}{b}$

Then $f(3^+) = 2 \lim_{x \rightarrow 3^+} \left(\frac{\sin(x-3)}{x-3} \right) = 2$ and

$f(3) = b.$

Hence $f(3) = f(3^+) = f(3^-)$

$$\Rightarrow b = 2 = -\frac{a}{b}$$

$b = 2, a = -4$

Hence only 1 ordered pair $(-4, 2)$

120. Consider the function $f : (0, 2) \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{2} + \frac{2}{x}$ and the function $g(x)$ defined by

$$g(x) = \begin{cases} \min(f(t)), & 0 < t \leq x \text{ and } 0 < x \leq 1 \\ \frac{3}{2} + x, & 1 < x < 2 \end{cases} \quad . \text{ Then :}$$

(A) g is continuous but not differentiable at $x=1$ (B) g is continuous and differentiable for all $x \in (0, 2)$

(C) g is not continuous for all $x \in (0, 2)$ (D) g is neither continuous nor differentiable at $x = 1$

$$f(x) = \frac{x}{2} + \frac{2}{x} \text{ द्वारा परिभाषित फलन } f : (0, 2) \rightarrow \mathbb{R} \text{ तथा } g(x) = \begin{cases} \min(f(t)), & 0 < t \leq x \text{ and } 0 < x \leq 1 \\ \frac{3}{2} + x, & 1 < x < 2 \end{cases} \text{ द्वारा परिभाषित}$$

फलन $g(x)$ का विचार कीजिए। तो :

(A) $x = 1$ पर g संतत है परन्तु अवकलनीय नहीं है। (B) सभी $x \in (0, 2)$ के लिए g संतत तथा अवकलनीय है।

(C) सभी $x \in (0, 2)$ के लिए g संतत नहीं है। (D) $x = 1$ पर g न तो संतत है न ही अवकलनीय है।

Ans. A

Sol. $f(x) = \frac{x}{2} + \frac{2}{x}$

$$\Rightarrow f'(x) = \frac{1}{2} - \frac{2}{x^2} = \left(\frac{x^2 - 4}{2x^2} \right) \text{ for } x \in (0, 2)$$

$$\Rightarrow f(x) < 0 \quad \Rightarrow f(x) \text{ is decreasing}$$

$$g(x) = \begin{cases} \min\{f(t)\} & , 0 < t \leq x \text{ and } 0 < x \leq 1 \\ \frac{3}{2} + x & , 1 < x < 2 \end{cases}$$

Since $f(x)$ is decreasing, $f(t)$ is minimum for x in $t \in (0, x] \Rightarrow \min\{f(t)\} = f(x)$

$$\Rightarrow g(x) = \begin{cases} f(x) & , x \in (0, 1] \\ \frac{3}{2} + x & , 1 < x < 2 \end{cases}$$

$$\Rightarrow g(1) = \frac{1}{2} + \frac{2}{1} = \frac{5}{2}$$

$$\text{Also } g(1^+) = \frac{3}{2} + 1 = \frac{5}{2}$$

$$\Rightarrow g(x) \text{ is continuous in } (0, 2)$$

$$g'(x) = \begin{cases} f'(x) & , x \in (0, 1] \\ 1 & , x \in (1, 2) \end{cases}$$

$$g'(1) = f'(1) = \frac{1-4}{2} = \frac{-3}{2}$$

$$\Rightarrow g'(1) \neq g'(1^+) \Rightarrow \text{not differentiable at } x = 1$$